



Design and Analysis of Algorithms

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DESIGN AND ANALYSIS OF ALGORITHMS

Method of Limits for comparing order of Growth

Slides courtesy of **Anany Levitin**

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Using Limits to Compare Order of Growth



$$\lim_{n \rightarrow \infty} \frac{t(n)}{g(n)} = \begin{cases} 0 & \text{implies that } t(n) \text{ has a smaller order of growth than } g(n), \\ c & \text{implies that } t(n) \text{ has the same order of growth as } g(n), \\ \infty & \text{implies that } t(n) \text{ has a larger order of growth than } g(n). \end{cases}$$

Case1: $t(n) \in O(g(n))$

Case2: $t(n) \in \Theta(g(n))$

Case3: $g(n) \in O(t(n))$

$t'(n)$ and $g'(n)$ are first-order derivatives of $t(n)$ and $g(n)$

L'Hopital's Rule $\lim_{n \rightarrow \infty} \frac{t(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{t'(n)}{g'(n)}$

Stirling's Formula $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$ for large values of n

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Using Limits to Compare Order of Growth: Example 1



Compare the order of growth of $f(n)$ and $g(n)$ using method of limits

$$t(n) = 5n^3 + 6n + 2, \quad g(n) = n^4$$

$$\lim_{n \rightarrow \infty} \frac{t(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{5n^3 + 6n + 2}{n^4} = \lim_{n \rightarrow \infty} \left(\frac{5}{n} + \frac{6}{n^3} + \frac{2}{n^4} \right) = 0$$

As per case1

$$t(n) = O(g(n))$$

$$5n^3 + 6n + 2 = O(n^4)$$

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Using Limits to Compare Order of Growth: Example 2



$$t(n) = \sqrt{5n^2 + 4n + 2}$$

using the Limits approach determine $g(n)$ such that $f(n) = \Theta(g(n))$

Leading term in square root n^2

$$g(n) = \sqrt{n^2} = n$$

$$\lim_{n \rightarrow \infty} \frac{t(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\sqrt{5n^2 + 4n + 2}}{\sqrt{n^2}}$$

$$= \lim_{n \rightarrow \infty} \sqrt{\frac{5n^2 + 4n + 2}{n^2}} = \lim_{n \rightarrow \infty} \sqrt{5 + \frac{4}{n} + \frac{2}{n^2}} = \sqrt{5}$$

non-zero constant

Hence, $t(n) = \Theta(g(n)) = \Theta(n)$

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Using Limits to Compare Order of Growth



$$\lim_{n \rightarrow \infty} t(n)/g(n) \neq 0, \infty \Rightarrow t(n) \in \Theta(g(n))$$

$$\lim_{n \rightarrow \infty} t(n)/g(n) \neq \infty \Rightarrow t(n) \in O(g(n))$$

$$\lim_{n \rightarrow \infty} t(n)/g(n) \neq 0 \Rightarrow t(n) \in \Omega(g(n))$$

$$\lim_{n \rightarrow \infty} t(n)/g(n) = 0 \Rightarrow t(n) \in o(g(n))$$

$$\lim_{n \rightarrow \infty} t(n)/g(n) = \infty \Rightarrow t(n) \in \omega(g(n))$$

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Using Limits to Compare Order of Growth: Example 3



Compare the order of growth of $t(n)$ and $g(n)$ using method of limits

$$t(n) = \log_2 n, g(n) = \sqrt{n}$$

$$\lim_{n \rightarrow \infty} \frac{\log_2 n}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{(\log_2 n)'}{(\sqrt{n})'} = \lim_{n \rightarrow \infty} \frac{(\log_2 e) \frac{1}{n}}{\frac{1}{2\sqrt{n}}} = 2 \log_2 e \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$$\log_2 n \in o(\sqrt{n})$$

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Orders of growth of some important functions



- All logarithmic functions $\log_a n$ belong to the same class

$\Theta(\log n)$ no matter what the logarithm's base $a > 1$ is

$$\log_{10} n \in \Theta(\log_2 n)$$

- All polynomials of the same degree k belong to the same class:

$$a_k n^k + a_{k-1} n^{k-1} + \dots + a_0 \in \Theta(n^k)$$

- Exponential functions have different orders of growth for different a 's

$$3^n \notin \Theta(2^n)$$

- order $\log n < \text{order } n^\alpha \ (\alpha > 0) < \text{order } a^n < \text{order } n! < \text{order } n^n$

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How to Establish Orders of Growth of an Algorithm's Basic Operation Count



Summary

- Method 1: Using limits.
 - L' Hôpital's rule
- Method 2: Using the theorem.
- Method 3: Using the definitions of O -, Ω -, and Θ -notation.



THANK YOU

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